

PALAMAKULA DEEPIKA

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PROFESSIONAL SUMMARY

A Computer Science graduate with a strong foundation in software development, skilled in Python and SQL, with the ability to build efficient, scalable solutions. Experienced in data analysis and Machine Learning tools to extract insights and support decision-making, along with improving system efficiency through structured coding and automation. A strong collaborator in agile environments, focused on delivering reliable solutions and continuous improvement.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C.

Machine Learning & AI: Scikit-learn, Pandas, NumPy, TensorFlow, Random Forest, Natural Language Processing (NLP).

Cloud & DevOps: AWS (EC2, S3), Git, GitHub, CI/CD (Basics), Docker (Basics).

Data Visualization & BI: Power BI, Grafana, Microsoft Excel, Exploratory Data Analysis (EDA).

Frontend & Design: HTML, CSS, Figma, UI/UX Design.

Core Competencies: Data Structures & Algorithms (Intermediate), Analytical Problem Solving, Cross-functional Collaboration, Agile Development.

WORK EXPERIENCE

Machine Learning Intern | FabHost Complete Web Solutions

Jan 2024 – Mar 2024

- Developed and deployed supervised machine learning models using Python and XGBoost, achieving 92% precision-recall score, applied SHAP (Explainable AI) for model interpretability and feature importance analysis.
- Architected end-to-end data pipelines including data preprocessing, feature engineering, and model validation, improving model robustness across datasets of 10,000+ records.
- Performed hyperparameter tuning to optimize model performance and enhance prediction accuracy. Automated data cleaning, preprocessing, feature selection, and data transformation using Python, reducing manual effort by 30% and improving workflow efficiency.

PROJECTS

CareerLens AI: Scalable Career Recommendation System | Nov 2025 – Jan 2026

- Developed an AI-based career recommendation system using Random Forest and XGBoost, achieving 90%+ accuracy through data preprocessing, feature engineering, and hyperparameter tuning on 10,000+ records.
- Designed a scalable 3-tier architecture with a Flutter frontend, FastAPI backend, and ML models, enabling real-time predictions via RESTful APIs. Implemented Explainable AI (SHAP) to provide transparent recommendations with feature importance and confidence scores.
- Built and managed an end-to-end ML pipeline including data ingestion, model training, evaluation, and deployment. Deployed the application on AWS (EC2, S3) with MongoDB, ensuring scalable, cloud-based model hosting and efficient data handling.

HapticNav: Assistive Navigation System with Android Monitoring | Jun 2024 – Aug 2024

- Developed an end-to-end assistive navigation system using ESP32, ultrasonic sensors, and GPS for real-time obstacle detection and location tracking. Implemented spatial haptic feedback mechanisms to provide directional guidance (left, right, stop), reducing reliance on audio and improving accessibility.
- Built a companion Android application for navigation control, live location tracking, and session history management.
- Integrated wireless communication (Bluetooth/WiFi) between device and app for seamless data exchange and event-driven alerts. Enabled offline navigation using OpenStreetMap with a focus on low-latency edge processing and system optimization for reliable performance.

EDUCATION

Bachelor of Technology | Artificial Intelligence & Machine Learning

CGPA: 8.2 / 10.0

Saveetha Engineering College, Chennai, Tamil Nadu, India | 2021 – 2025

CERTIFICATIONS

- AWS Certified Cloud Practitioner** — Amazon Web Services (AWS -Udemy), 2025
- Fundamentals of DevOps** — Simplilearn, 2024

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